

Final Exam – Databases 2015

March 24th, 2015

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General information:

- The exam consists of 4 pages and 5 tasks.
- The maximal number of attainable points is 70.
- The grade for this exam is the minimum of 10 and $(1 + \frac{9}{70} \cdot \text{your points})$.
- **Write in clear handwriting and if possible answer in English.**

1. Conceptual Design

Assume the following scenario. You have gotten a job at a large international company. Your first job assignment is to help maintain information about the company and its employees. Your model should capture the following information:

- The company has different branches, each having an address and a unique name. Every branch consists of one or more departments. The name of the departments is not unique, but there cannot be two departments with the same name in a single branch.
- Every employee has a first-name, last-name, id, and birthday. An employee may also have a middle name.
- Every department has a department head.
- Every employee works for one or more departments (possibly at different branches). Each of these work assignments has a number of hours per week and a salary assigned to it. Moreover, each work assignment may be supervised by another employee. (Note that the supervisor is not necessarily the boss and different work assignments of the same employee may have different supervisors.)

(a) Provide a conceptual database model in the form of an E/R diagram. **(6 points)**

(b) Give the associated relational schema. Design your relational schema such that the cardinality constraints in your E/R diagram are captured (enforced) as good as possible.

Indicate relations $R(A_1, A_2, \dots)$ with their attributes $A_1, A_2, \dots$, primary keys underlined, and foreign key relations $(\rightarrow)$.

Comment also on NULLable attributes, and constraints that a relational database could check, including candidate keys. **(4 points)**

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2. Functional Dependencies

Given is a relation $R(A, B, C, D)$ and following functional dependencies:

$$\begin{array}{ll} A, B \rightarrow C, D & A, C \rightarrow D \\ C \rightarrow A, D & A, C, D \rightarrow B \\ B, C, D \rightarrow A & \end{array}$$

- (a) Determine a canonical (minimal) basis for the functional dependencies. **(4 points)**
- (b) What are the minimal keys? **(3 points)**
- (c) What does it mean that a functional dependency α is implied by a set of functional dependencies Φ ? **(2 points)**
- (d) Do the functional dependencies above imply $C \rightarrow B, D$? **(1 points)**

3. Normalisation

Given is a relation $R(A, B, C, D, E, F)$ and the following **canonical** functional dependencies:

$$\begin{array}{ll} C, E \rightarrow B & D \rightarrow B \\ D \rightarrow E & A \rightarrow D \\ D, F \rightarrow A & C, E \rightarrow F \end{array}$$

Show your **intermediate steps** in all the answers.

- (a) Describe *two* anomalies that can occur if a schema violates a normal form. **(2 points)**
- (b) Transform R to *Boyce-Codd normal form* using the synthesis algorithm for BCNF. **(4 points)**
- (c) Transform R to *third normal form* using the synthesis algorithm for 3NF. **(3 points)**
- (d) Which functional dependencies have been lost in transformations (b) or (c)? **(1 points)**

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4. Structured Query Language

Consider the following schema:

- Customer(cid, cname)
The customers with an identification number and name.
- Product(pid, pname, category , price)
The products with an identification number, name, category and price.
- Order(orderNr, date, cid $\rightarrow$ Customer, pid $\rightarrow$ Product)
*The orders with an order number, a date and references to customer and product.
(For simplicity, we assume that an order contains precisely one product.)*

Formulate the following queries in SQL.

- (a) Find the name of all customers that have placed **at least two** orders. (5 points)
- (b) Find the name of all customers that have placed **no** orders. (5 points)
- (c) Find the name of all products that have been ordered **less than 5** times. (5 points)
- (d) Find the name of all customers that have ordered all products in the category *books*. (5 points)

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5. Transactions

- (a) Determine the precedence graph for the following schedule and decide whether the schedule is conflict-serialisable: **(5 points)**

T1	R(B)	W(B)	W(A)
T2		R(A)	W(C)
T3	R(B)		W(C)

- (b) Show that the following schedule is realisable with two-phase locking. For this purpose, insert all necessary lock and unlock actions. **(6 points)**

T1	R(B)	W(A)
T2	R(A)	W(C)
T2		R(C)

- (c) Is the schedule in item (b) realisable with **(2 points)**
- strict two-phase locking, or
 - preclaiming two-phase locking?
- (d) Give an example of a schedule that is realisable with strict two-phase locking but *not* with preclaiming two-phase locking. You may use *start* and *commit* to indicate in the schedule when transactions are started and committed. **(2 points)**
- (e) Name and briefly describe the three phases of optimistic concurrency control. For the case of backward-oriented concurrency control also describe what the database checks when a transaction tries to commit (the condition for successful commit). **(5 points)**